

Probability Distributions and Statistical Analysis

1. Objective

To study and apply important probability distributions using Python, with emphasis on Bernoulli, Binomial, Poisson, Normal, and Uniform distributions. The experiment calculates probabilities, mean, variance, percentiles and distribution characteristics, and verifies the results using SciPy statistical functions and visualizations.

2. Description

- **Bernoulli Distribution:** Models a single trial with two outcomes, such as a library gate card being read successfully or not.
- **Binomial Distribution:** Models the number of correct answers in a fixed number of independent multiple-choice guesses.
- **Poisson Distribution:** Models the number of events occurring in a fixed time interval, such as help-desk emails or server errors.
- **Normal Distribution:** Used for continuous measurements such as phone battery life and coffee-machine fill volume.
- **Uniform Distribution:** Models equally likely waiting times, such as a campus shuttle arriving between 0 and 15 minutes.
- **Python verification:** SciPy functions are used to calculate PMFs, CDFs, survival probabilities, percentiles, means and variances.
- **Visualization:** Discrete distributions are represented using bar plots, while continuous distributions are represented using density curves.

3. Model / Flowchart / Architecture



Generate PMF / PDF visualizations



Interpret results



4. Algorithm

1. START
2. Import the required Python libraries: NumPy, Matplotlib and SciPy.
3. Define the parameters for each probability distribution.
4. For Bernoulli, calculate $P(X=0)$, $P(X=1)$, mean and variance.
5. Estimate Bernoulli p from the attendance data and compare theoretical and simulated relative frequencies.
6. For Binomial, verify the BINS conditions and calculate the probability of exactly 5 correct answers out of 12.
7. For Poisson, calculate probabilities for specified event counts and evaluate the probability of exceeding a threshold.
8. For Normal distribution, calculate Z-scores, use the 68–95–99.7 rule, estimate an unknown standard deviation and calculate percentiles.
9. For Uniform distribution, calculate mean and variance for the shuttle waiting-time interval.
10. Use scipy.stats functions to verify the hand calculations.
11. Generate suitable plots for discrete and continuous distributions.
12. Round reported probabilities to four decimal places.
13. Interpret the results and END the experiment.

Important Formulae Used

Distribution	Formula / Result
Bernoulli	$P(X=x)=p^x(1-p)^{1-x}$; $E[X]=p$; $\text{Var}(X)=p(1-p)$
Binomial	$P(X=k)=C(n,k)p^k(1-p)^{n-k}$; $E[X]=np$; $\text{Var}(X)=np(1-p)$
Poisson	$P(X=k)=e^{-\lambda}\lambda^k/k!$; $E[X]=\text{Var}(X)=\lambda$
Normal	$Z=(X-\mu)/\sigma$
Uniform	$E[X]=(a+b)/2$; $\text{Var}(X)=(b-a)^2/12$

5. Selected Results / Observations

Experiment	Result / Observation
Q1 – Bernoulli	$p = 0.92$; $P(X=1) = 0.9200$; $P(X=0) = 0.0800$; mean = 0.9200; variance = 0.0736.
Q2 – Bernoulli estimation	1740 accepted scans out of 2000 gives $p = 0.8700$; variance = 0.1131. Bernoulli variance is maximum at $p = 0.5$, with maximum variance 0.25.
Q3 – Binomial	For $n = 12$ and $p = 0.25$, $P(X=5) \approx 0.1032$; mean = 3.0000; variance = 2.2500. BINS conditions are satisfied.
Q5 – Poisson	For $\lambda = 3$, $P(X=0) \approx 0.0498$ and $P(X=2) \approx 0.2240$. Mean and variance are both 3.

Q6 – Poisson	For 8 hours at 1.5 errors/hour, $\lambda = 12$. $P(X=0) \approx 0.0000$ to four decimals; $P(X>18) \approx 0.0374$.
Q7 – Normal	For $\mu = 18$ and $\sigma = 1.5$, $X = 15$ gives $Z = -2$. Approximately 95% of values lie between 15 and 21 hours.
Q8 – Normal	For $\mu = 200$ and $P(X>210)=0.025$, $\sigma \approx 5.1020$ ml. The 1st percentile is ≈ 188.13 ml and $P(X<190) \approx 0.0250$.
Q9 – Uniform	For waiting time $U(0,15)$, mean = 7.5000 minutes and variance = 18.7500 minutes ² .

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5. Selected Results / Interpretation (continued)

The Bernoulli examples demonstrate how a binary outcome can be represented by a single probability p . The attendance example also shows how p can be estimated directly from observed successes and total trials. The variance $p(1-p)$ reaches its maximum at $p=0.5$.

The Binomial example extends the Bernoulli model to a fixed number of independent trials. With four equally likely answer choices, the probability of a correct random guess is 0.25, and the expected number of correct answers in 12 questions is 3.

The Poisson examples model event counts over time. For a Poisson distribution, the mean and variance are equal to λ . Increasing the observation period changes the rate parameter proportionally; therefore, the 1.5 errors/hour rate becomes $\lambda=12$ over an 8-hour shift.

The Normal examples demonstrate standardization through Z-scores, percentile calculations, and the 68–95–99.7 rule. A value of 15 hours with mean 18 hours and standard deviation 1.5 hours is two standard deviations below the mean.

The Uniform example demonstrates a continuous distribution in which every value in the specified interval has equal density. For $U(0,15)$, the midpoint is 7.5 minutes and the variance is 18.75 minutes².

6. Visualization

Distribution	Visualization	Purpose
Bernoulli	Bar plot / PMF	Compare probabilities of the two outcomes.
Binomial	Bar plot / PMF	Show probabilities for 0 through 12 correct answers.
Poisson	Bar plot / PMF	Show event-count probabilities for a selected time interval.
Normal	Density curve / PDF	Show the continuous bell-shaped probability density.
Uniform	Density curve / PDF	Show constant density over the waiting-time interval.

7. GitHub Link

https://github.com/vivekkhenibtech25/Data_sci_lab/tree/main/lab4.b

8. Conclusion

The experiment demonstrates the application of five fundamental probability distributions using Python and SciPy. Bernoulli and Binomial distributions model binary and repeated fixed-trial experiments, Poisson models event counts over time, Normal models continuous measurements using mean and standard deviation, and Uniform models equally likely values over an interval. The hand calculations were verified using `scipy.stats`, while probability mass functions and density functions were visualized to support interpretation. The exercise provides a practical foundation for selecting an appropriate probability model and interpreting its statistical parameters.

EXPERIMENT COMPLETED